

Final Project

■ Team:

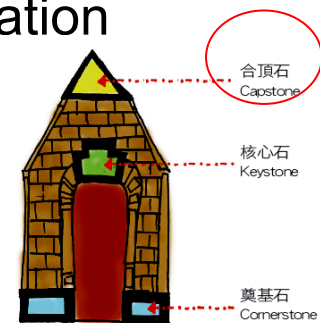
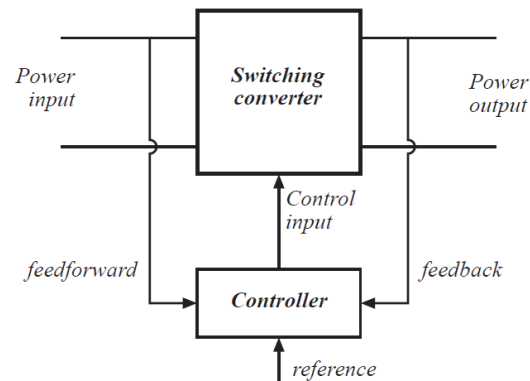
- 3~4 persons, different background is better

■ Software:

- SIMPLIS, SPICE or other IC design tools

■ Topics:

- 利用上課所學完成切換式電源及其控制器的設計 (close-loop), 並用電腦軟體模擬. 架構自選 (spec./applications, power stage, control method, focus points)
- At least **three blocks** implemented in MOS level (Any fabrication process is acceptable)
- Other blocks could be implemented by behavioral model
- Apply what you learned this semester to this design project



Progress Presentation Schedule

5/7	Project plan and topics
5/14	Design spec., Behavior model simulation
5/21	Behavior model simulation: transient + AC simulation
5/28	IC subblock simulation
6/4	IC subblock simulation (2)
6/11	Final presentation

- Progress presentation per team: ~5 minutes + 1 minutes Q&A



Required Content of Progress Presentation

- Project plan and topics
 - Topic, applications, motivation, literature survey
 - Expected MOS-level-design subblocks, fabrication process, simulation tools
 - System spec.
- Design spec., Behavior model simulation
 - Specifications for power stage and controller (obtained from system spec.)
 - Use simulation to show that design meets specifications
 - Steady-state, start-up, transient, AC



Required Content of Progress Presentation (2)

- IC subblock design and simulation
 - Specifications for subblocks (obtained from controller spec.)
 - Use simulation to show that design meets specifications
 - (Optional) Whole chip close-loop simulation: All MOS level blocks or (MOS level blocks + behavior model blocks)



Final Project (2)

■ Grading:

- Progress presentations: 20% (from professor)
- Final presentation: 30% from students, 50% from professor
- Additional bonus points will be given to those students who ask questions
- Honor code: Please do not use the materials from previous years

■ PPT format

- First page: title, name (group members and presenters), date
- Second page: outline
- Main part

- Conclusions and references



Final Presentation

- Presentation time: 15 minutes + 2 minutes Q&A
- 內容: 之前進度的總結
 - Topic
 - Introduction
 - Design Spec.
 - Behavior analysis and simulation
 - IC subblock analysis and simulation
 - (Optional) Whole chip close-loop simulation (MOS level blocks + behavior model blocks)
 - Conclusions
 - References



Honor Code

- **Do not use the materials from previous years.**

